Universität Basel

PROPOSAL

Crypto Liquidity and Global Risk Appetite

Evidence from Stablecoin Issuance and Other Indicators

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52355-01 – Blockchain, Smart Contracts & DeFi
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Submitted: 15-Mar-2026

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1 Introduction and Research Question

Over the past decade, crypto markets have evolved from a niche segment into a visible part of the global financial system. Within this ecosystem, stablecoins have become especially important because they provide a relatively stable medium for trading, settlement, and liquidity provision. Recent research shows that stablecoin stability depends heavily on arbitrage and market design (Lyons & Viswanath-Natraj, 2023), which underlines their role as core liquidity infrastructure rather than merely passive payment tokens. Stablecoins may therefore offer useful information about changes in risk-taking within crypto markets.

At the same time, the literature suggests that major cryptocurrencies behave less like traditional money and more like speculative assets (Baur, et al., 2018). Bitcoin, in particular, is found to be used mainly as an investment rather than as a medium of exchange. Other studies further show that crypto markets are often relatively separate from traditional asset markets, though linkages can strengthen during periods of stress and rapid price movements (Corbet, et al., 2020). This raises the possibility that changes in crypto liquidity may reflect broader swings in investor sentiment.

Against this background, this paper asks whether stablecoin issuance and other crypto liquidity indicators can help predict global risk appetite. The aim is not to identify a permanent trading rule, but to examine whether developments in crypto markets can contain information about future movements in equity returns, volatility, and credit conditions. This idea is consistent with the broader finance literature, which shows that sentiment can shape the pricing of speculative assets and that predictive relationships should be interpreted with caution.

Finally, the paper will contribute by examining whether stablecoin issuance and related crypto-liquidity indicators contain predictive information about broader market-based measures of risk appetite.

2 Related Literature

The existing literature on this topic can be grouped into three related strands. The first examines stablecoins and liquidity within crypto markets, the second studies the link between crypto assets and traditional financial markets, and the third provides the broader empirical finance framework on predictive relationships.

2.1 Stablecoins and crypto market liquidity

A first strand of the literature studies stablecoins and their role within crypto markets. Lyons and Viswanath-Natraj examine what keeps stablecoins close to their peg and show that arbitrage and redemption mechanisms are central for stability (Lyons & Viswanath-Natraj, 2023). This highlights the importance of stablecoins for trading and settlement infrastructure. Makarov and Schoar show that crypto markets are fragmented across exchanges and that limited arbitrage allows price differences to persist, making liquidity conditions especially important (Makarov & Schoar, 2020). Griffin and Shams add that Tether issuance is associated with subsequent Bitcoin price increases, suggesting that stablecoin flows may affect market liquidity and price dynamics more broadly (Griffin & Shams, 2020).

2.2 Crypto assets and broader financial markets

A second strand asks how cryptocurrencies relate to traditional financial assets and investor risk-taking. Baur, Hong, and Lee find that Bitcoin behaves primarily as a speculative asset rather than as a true medium of exchange, which makes it relevant as an indicator of risk appetite (Baur, et al., 2018). Corbet et al. show that cryptocurrencies are often weakly connected to traditional assets in normal periods, although these links may strengthen during market stress (Corbet, et al., 2020). This suggests that crypto markets may at times reflect broader changes in financial conditions and sentiment, even if they do not move in lockstep with traditional markets.

2.3 Predictive factors, sentiment, and financial conditions

A third strand provides the broader empirical finance background for this research. Baker and Wurgler show that investor sentiment matters for asset prices, especially for more speculative assets (Baker & Wurgler, 2006). Campbell and Thompson discuss predictive regressions in finance and stress that forecasting relationships often look stronger in-sample than out-of-sample, which is important for interpreting predictive results with caution (Campbell & Thompson, 2005). Adrian, Boyarchenko, and Giannone further show that financial conditions contain information about broader macro-financial vulnerability (Adrian, et al., 2019). Taken together, this literature supports the idea that market-based indicators can contain useful information about future risk conditions, while also warning that such relationships may not be stable over time.

3 Data

This study will primarily use daily data, as this frequency is well suited to capturing the high volatility and rapid adjustment of crypto markets while still allowing for meaningful comparison with traditional financial indicators. The sample period will depend on data availability but is expected to begin once stablecoin markets became economically significant and reliable data coverage is available, likely in the late 2010s. Using daily observations over a sufficiently long horizon should make it possible to capture both normal market conditions and episodes of financial stress, which is important for assessing whether crypto liquidity indicators contain information about broader shifts in global risk appetite.

3.1 Crypto Liquidity Indicators

The main measure of crypto liquidity in this paper will be stablecoin supply growth. This variable is most closely aligned with the core research question, as it captures expansions in liquidity that can be used directly for trading, settlement, and collateral within crypto markets. In this sense, it provides a broad and intuitive indicator of whether risk-taking capacity in the crypto ecosystem is increasing. Additional proxies

such as futures open interest, funding rates, and trading volume can complement this measure by capturing more specific aspects of market activity, including leverage demand, speculative positioning, and overall trading intensity. Including these variables therefore helps provide a more complete picture of crypto liquidity and allows the analysis to assess which indicator contains the strongest information about broader risk sentiment.

3.2 Global Risk Sentiment Measures

For global risk sentiment, this paper will mainly rely on equity market returns, the VIX, and credit spreads. These measures capture three complementary dimensions of market sentiment and financial conditions: the performance of risky assets, the level of market uncertainty, and the pricing of default risk in credit markets. Together, they provide a broad and intuitive picture of whether financial conditions are improving or deteriorating. This choice is also consistent with the broader empirical finance literature, which shows that sentiment and financial conditions play an important role in shaping asset prices and macro-financial vulnerability. Additional proxies such as a financial conditions index, high-yield spreads, or other sentiment-related market indicators may also be useful in robustness tests, as they can capture more specific aspects of investor optimism and risk-taking. Including these alternative measures therefore helps ensure that the results are not driven by any single definition of global risk sentiment.

4 Empirical Methodology

4.1 Predictive Regressions

Predictive regressions provide a simple way to test whether current crypto liquidity conditions contain information about future developments in broader financial markets. In this study, they will be used to examine whether changes in stablecoin supply growth and related liquidity proxies help predict subsequent movements in equity returns, the VIX, and credit spreads. The basic idea is to relate future measures of global risk sentiment to today's crypto liquidity indicators, while also controlling for other

market variables where appropriate. This approach is well suited to the research question because it allows the analysis to assess whether crypto market developments tend to lead, rather than merely coincide with, changes in broader financial conditions.

More specifically, the regression framework can be used to compare the predictive content of different liquidity measures, such as stablecoin issuance, open interest, funding rates, and trading volume. This makes it possible to identify which indicator appears most informative for future risk sentiment. At the same time, the results must be interpreted carefully, since the empirical finance literature shows that predictive relationships are often unstable over time and may weaken once they become widely known.

4.2 Vector Autoregression (VAR)

A vector autoregression (VAR) will complement the predictive regressions by taking a broader view of the relationship between crypto liquidity and global risk sentiment. Rather than focusing on a single forecasting equation, this approach treats the main variables as evolving jointly over time. In the context of this paper, this means that stablecoin issuance, crypto returns, the VIX, equity returns, and credit spreads can all influence one another. A VAR therefore makes it possible to move beyond the question of whether stablecoin issuance predicts, for example, the VIX and to also consider whether the VIX may in turn affect stablecoin issuance, whether crypto returns influence both, and whether these variables interact over several periods.

This is particularly useful for the present research question because the relationship between crypto markets and broader financial conditions is unlikely to run in only one direction. The VAR framework can therefore help identify broader transmission patterns and potential feedback effects between crypto liquidity and traditional market sentiment. In this sense, it serves as a natural second step after the predictive regressions: while the predictive regressions test whether crypto liquidity indicators contain information about future movements in global risk sentiment, the VAR helps examine the dynamic interactions through which such relationships may emerge.

5 Preliminary Paper Structure

1. Introduction

Introduction to the research question, motivation, and main contribution of the paper.

2. Related Literature

Review of the literature on stablecoins and crypto market liquidity, the relation between crypto assets and traditional financial markets, and the broader literature on sentiment, financial conditions, and predictive relationships.

3. Data and Variable Construction

Presentation of the data sources, the construction of the main crypto liquidity indicators, and the definition of the global risk sentiment measures used in the analysis.

4. Empirical Methodology

Description of the predictive regression framework and the vector autoregression (VAR) approach used to study both forecasting relationships and broader dynamic interactions.

5. Empirical Results

Presentation of the main findings on whether crypto liquidity indicators help predict changes in global risk sentiment and which proxies appear most informative.

6. Robustness Tests and Extensions

Assessment of whether the results remain stable when alternative liquidity proxies, additional sentiment measures, or different sample specifications are used.

7. Discussion

Interpretation of the findings considering the existing literature and discussion of what they imply for the role of crypto markets in broader financial conditions.

8. Conclusion

Summary of the main results, contribution, limitations, and possible directions for future research.

6 Data Sources

For financial market data, this paper will primarily rely on the Bloomberg Terminal, as it provides convenient access to a broad range of variables through Excel. If needed, additional or alternative sources such as FRED, Yahoo Finance, or LSEG/Refinitiv may also be used.

For crypto market data, the analysis will mainly rely on DeFiLlama and CoinGecko. At the same time, other crypto data providers may also be considered should a more suitable source for specific liquidity indicators become necessary.

References

The reference list includes both cited literature and additional sources relevant to the planned analysis.

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Data sources

Bloomberg Finance L.P. *Bloomberg Terminal*.

CoinGecko. *Cryptocurrency market data platform*.

DeFiLlama. *Stablecoin and DeFi data platform*.

Federal Reserve Bank of St. Louis. *Federal Reserve Economic Data (FRED)*.

LSEG / Refinitiv. *Financial market data platform*.